Aggregation determines scaling: power-law verdicts in dengue surveillance data are conditional on the spatio-temporal window

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November 3, 2026

Abstract

Power-law scaling is routinely reported for infectious disease incidence, and dengue in Brazil is a standard example: heavy-tailed distributions of case counts are invoked to argue that extreme epidemics are structurally unpredictable. We show that the verdict depends less on dengue than on the distribution one chooses to build. Using weekly dengue surveillance data spanning January 2010 to January 2026 for the 5,571 Brazilian municipalities, processed through the Mosqlimate/Infodengue platform, we fitted discrete power laws by maximum likelihood with Kolmogorov–Smirnov $x_{\min}$ selection at seven nested aggregation levels and compared each fit against an exponential alternative by a likelihood-ratio test. The results split sharply. For the cross-sectional distribution of *epidemic size* across municipalities within a year, the power law is preferred in 100% of the 15 national fits (all significant at $p < 0.05$; $\alpha \approx 2.0$) and in 96% of 330 state-year fits, with $x_{\min}$ in the hundreds of cases. For the distribution of *epidemic duration*, the power law is rejected in all 15 national fits (and in 95% of the state-year fits) in favour of an exponential, with α piling up at 2.90–2.96. For weekly case counts, the power law appears preferred in 75% of city-level series but $x_{\min} \leq 5$ in 77% of them, meaning that the fitted object is the bulk of the distribution rather than a tail. We conclude that the heavy-tail literature on dengue conflates at least three distinct random variables, that scale claims are only interpretable where $x_{\min}$ is far above the bulk, and that reporting α without the (unit, $x_{\min}$, comparison model) tuple is uninformative.

1 Introduction

Understanding the dynamics of infectious disease epidemics has led to a very prolific literature, both in terms of mechanistic mathematical models and of statistical models (Siettos and Russo, 2013). The former attempt to capture the causal chain of events involved in disease transmission, the latter try to make sense of a growing collection of data about infectious disease incidence.

Among the most salient results of the mathematical modelling of infectious diseases was the definition of the basic reproductive number, R_0, which determines the most important characteristics of epidemics such as their spreading speed and the epidemic threshold, also known as the herd immunity threshold. R_0 appears in the equations governing prevalence growth as an exponent determining the initial exponential growth

of an epidemic, and it defines an important bifurcation point at $R_0 = 1$ linked to the existence and stability of equilibria in epidemic models.

On the statistical side, the largest portion of the literature deals with statistical modelling of incidence data, i.e. the number of cases accumulated over time and space or studied as a stochastic process, such as a Poisson or negative binomial process (Becker and Britton, 1999; Li et al., 2012). The central difficulty is overdispersion: real surveillance counts produce far more extreme observations than any tractable count model comfortably allows.

A recurrent claim is that this overdispersion is not merely large but *scale-free*, i.e. that incidence follows a power law (Rhodes et al., 1997; Clauset et al., 2009; **?**). The attraction of this claim is that the very heavy tail implies extreme values that most models fail to predict — and those are precisely the events that cause the most disease and the largest economic burden. If true, this would have direct consequences for forecasting and for resource allocation.

The problem is that the empirical record is inconsistent. Reported exponents for dengue range widely, some studies find exponential rather than power-law behaviour, and the presence of a heavy tail appears to depend on the study period, the spatial unit and the outcome variable. Previous work has usually attributed this heterogeneity to biology or geography: different places, or different years, are assumed to be governed by different regimes.

Here we test a simpler and more uncomfortable explanation: that the heterogeneity is largely produced by the analyst. Any epidemiological time series can be cut into distributions in several legitimate ways — the weekly counts of one city, the annual totals of one city, the set of epidemic sizes across cities in one year, the set of epidemic durations — and these are different random variables with different supports and different tail behaviour. If the verdict changes as the construction changes, then a substantial part of the published disagreement is methodological rather than biological.

1.1 Contributions

This paper makes three contributions.

1. **A systematic seven-level comparison on one dataset.** We fit the same estimator to the same underlying surveillance data under seven nested aggregation levels (Table 1), so that any difference in the inferred regime is attributable to the construction and not to differing data sources, periods or software.

2. **An identification diagnostic that separates signal from artefact.** We show that the reported preference for a power law is only interpretable when the estimated lower bound $x_{\min}$ lies well above the bulk of the data. Where $x_{\min}$ collapses to 1–5 cases, the fit describes the entire distribution and the "tail" language is vacuous. We report $x_{\min}$ alongside every exponent and use it to classify each of the 47,494 fits.

3. **A resolved claim about dengue.** Epidemic *size* is heavy-tailed with $\alpha \approx 2$ and is stable across states and years; epidemic *duration* is not heavy-tailed and is better described by an exponential. The two variables are frequently conflated in the literature, which explains a large part of the apparent contradiction.

2 Methods

2.1 Data sources

Weekly dengue case counts were obtained from the InfoDengue/Mosqlimate platform (Araujo et al., 2025) for the period from 1 January 2010 to 22 January 2026 (the extraction date), for all Brazilian municipalities with available data. The variable used is the nowcast-adjusted estimated number of cases (`casos_est`), resampled to epidemiological weeks ending on Sunday (`W-SUN`); zero-case weeks are retained in the series but excluded from distributional fits, since the power-law family is defined on positive support.

Annual epidemic descriptors were obtained from the EpiScanner module of the same platform, which partitions each municipal series into epidemic episodes and reports, for each episode, its total size (*total cases*) and its duration in weeks (*ep_dur*). Annual cross-sections are available for the 15 epidemic years 2011–2025; the pooled all-years fit is stored separately (with a sentinel `year = 0`) and is excluded from all year counts. This yields, for each state-year and for each national year, a cross-section of municipal epidemics.

Municipal population figures were taken from the estimates distributed with the platform (`city_stats` table; 5,571 municipalities, one with a zero entry excluded) and used only for descriptive correlation. Population is not used in any fit.

2.2 Seven aggregation levels

The unit of analysis is the *set of positive observations* passed to the estimator. Table 1 defines the seven constructions used. The first three treat the individual municipality as the sampling unit and vary the temporal window; the last four treat a cross-section of municipalities within a period as the sampling unit.

Table 1: The seven aggregation levels. "Unit" is the sampling unit, i.e. what each observation in the fitted sample represents.

ID	Unit	Variable	Source table
L1	city	weekly cases, full series	`powerlaw_fits`
L2	city-year	weekly cases, one season	`powerlaw_fits_yearly`
L3	city	annual total, across years	`powerlaw_fits_total_cases`
L4	city $\subset$ state-year	epidemic size	`episcanner_state_fits`
L5	city $\subset$ state-year	epidemic duration	`episcanner_state_fits`
L6	city $\subset$ national-year	epidemic size	`episcanner_fits`
L7	city $\subset$ national-year	epidemic duration	`episcanner_fits`

Levels L1 and L2 differ only in the temporal window over which the weekly counts are pooled: L1 uses the complete available series for a municipality (January 2010 to January 2026), whereas L2 restricts to a single epidemic year (at most 52 weeks). Both sample sizes therefore differ by roughly an order of magnitude, which is why we compare shares rather than raw counts. Levels L4–L7 differ only in the spatial scope of the cross-section, not in the variable being measured. This nesting is deliberate: it allows each contrast in Section 3 to be read as the effect of a single change in construction.

2.3 Discrete power-law estimation

For each sample we fit a discrete power law by maximum likelihood following Clauset et al. (2009). The fitted density for integer $x \geq x_{\min}$ is

$$p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})}, \tag{1}$$

where $\zeta(\alpha, x_{\min})$ is the generalised (Hurwitz) zeta function, which normalises the discrete mass. The exponent α is estimated by numerical maximisation of the likelihood, and the lower bound $x_{\min}$ is selected jointly by minimising the Kolmogorov–Smirnov distance between the empirical and fitted cumulative distributions over all candidate values of $x_{\min}$. Because the data are counts, the discrete formulation is used throughout (`discrete=True`); the continuous approximation is not employed at any level.

The choice of $x_{\min}$ is not a nuisance parameter. It determines which fraction of the data the model actually describes, and therefore whether the fitted exponent refers to a tail at all. We therefore retain $x_{\min}$ for every fit and report its median and the proportion of fits with $x_{\min} \leq 5$ in each aggregation level.

2.4 Model comparison

A power law can always be fitted; the question is whether it describes the data better than a plausible alternative. We compare each power-law fit against an exponential distribution, which is the natural null for epidemiological counts because it corresponds to a constant hazard of termination and produces a light tail. The comparison follows the likelihood-ratio procedure of Clauset et al. (2009), as implemented in the `powerlaw` package with `normalized_ratio=True`.

$$R = \sum_{i=1}^{n} \left[\ln p_{\mathrm{PL}}(x_i) - \ln p_{\exp}(x_i) \right], \tag{2}$$

normalised so that its magnitude is comparable across samples of different size. The sign of R indicates the preferred model ($R > 0$ favours the power law, $R < 0$ favours the exponential) and its associated p-value quantifies whether the sign is distinguishable from sampling noise.

We adopt a deliberately conservative reporting convention. A level is described as *power-law supported* only when $R > 0$ and $p < 0.05$; as *exponential supported* only when $R < 0$ and $p < 0.05$; and as *inconclusive* otherwise. This prevents the common error of counting a positive but non-significant R as evidence for scaling.

2.5 Identification diagnostic

We flag a fit as *tail-informative* when $x_{\min}$ exceeds the bulk of the observed distribution, operationally $x_{\min} > 5$ cases for weekly counts and $x_{\min}$ above the median observed value for the aggregate levels. Where $x_{\min} \leq 5$, the fitted power law is being asked to describe the entire empirical distribution, including the small-count regime where the discrete power law is not a plausible generative model (case counts of 1 or 2 arise from reporting mechanics, not from scale-free branching). Such fits are retained in the database but are excluded from substantive interpretation, and we report the proportion explicitly.

2.6 Software and reproducibility

All fits are stored in a single SQLite database (`powerlaw_results.db`) committed to the project repository, together with the fitting code (`fitpl.py`) and the cached parquet extracts of every model result. Every number in this manuscript is derived from that database by the queries in Appendix A; no statistic is transcribed by hand. The full pipeline is therefore re-runnable, and a reader can confirm or contradict each figure with a single SQL statement.

3 Results

Table 2 summarises the 47,494 fits. The contrast between levels is the central result.

Table 2: Power-law fits by aggregation level. α median with interquartile range; $x_{\min}$ median; $\mathrm{PL}_{\mathrm{pref}}$ = proportion with $R > 0$; $\mathrm{PL}_{\mathrm{sig}}$ = proportion with $R > 0$ and $p < 0.05$; $\mathrm{Exp}_{\mathrm{sig}}$ = proportion with $R < 0$ and $p < 0.05$. Shares are percentages.

Level	n	α	IQR	$x_{\min}$	$\mathrm{PL}_{\mathrm{pref}}$	$\mathrm{PL}_{\mathrm{sig}}$	$\mathrm{Exp}_{\mathrm{sig}}$
L1 weekly, city	5,244	1.95	1.72–2.29	2	75	45	4
L2 weekly, city-year	40,616	2.06	1.72–2.45	2	23	3	21
L3 annual total, city	944	2.09	1.80–2.46	341	51	4	4
L4 size, state-year	330	1.97	1.79–2.14	159	96	52	0
L5 duration, state-year	330	2.80	2.52–2.92	17	0	0	95
L6 size, national-year	15	2.00	1.92–2.11	762	100	100	0
L7 duration, national-year	15	2.95	2.90–2.96	18	0	0	100

Epidemic size scales. At the national cross-section (L6), all 15 annual fits favour the power law and all are significant, with $\alpha = 2.00$ (IQR 1.92–2.11) and $x_{\min}$ of 762 cases. The exponent is remarkably stable: the annual values range only from 1.78 to 2.27. At state level (L4) the picture is consistent but weaker in significance, as expected from smaller cross-sections: the power law is preferred in 96% of 330 fits, significant in 52%, and α has a median of 1.97 across the 26 states (state-level medians spanning 1.70–2.58).

Duration does not scale. At the same cross-sectional levels, the duration of the same municipal epidemics behaves in the opposite way. At national level (L7) the power law is rejected in 15 of 15 years, with the exponential significant in every one. The estimated exponent piles up at 2.95 (range 2.90–2.96, SD 0.03), which is the signature of a truncated or exponential body being forced through a power-law estimator rather than evidence of scaling. At state level (L5) the exponential is significant in 95% of fits.

Weekly counts are not a tail. Levels L1 and L2 produce large apparent support for the power law (75% and, in the full-series case, 45% significant), but the estimated lower bound collapses to $x_{\min} = 2$ in the majority of fits: 77% of L1 and 81% of L2 fits have $x_{\min} \leq 5$. At that bound the model is fitted to essentially the whole distribution, including counts of 1 and 2 cases, and the heavy-tail interpretation does not follow. By contrast, in L3–L7 the median $x_{\min}$ is 17–762 cases, i.e. the fit describes genuine tails.

The window flips the verdict on the same data. The cleanest demonstration is the L1/L2 contrast: pooling the same weekly counts over one season instead of a decade drops the proportion of fits favouring the power law from 75% to 23% and raises the proportion significantly favouring the exponential from 4% to 21%. Nothing about dengue changed; only the window did.

3.1 Descriptive associations

The municipal exponent is essentially unrelated to city size: $\mathrm{corr}(\alpha, \log_{10}\text{population}) = -0.065$ (Spearman -0.072; $n = 5{,}244$). The association with incidence is weak and negative (Pearson -0.136, Spearman -0.248; $n = 4{,}663$). We therefore find no support for the view that larger or more heavily burdened municipalities sit in a different scaling regime; the exponent is a fairly constant property of the aggregation level.

4 Discussion

[To be written.]

5 Conclusion

[To be written.]

A Verification of reported quantities

Every quantity in Table 2 is obtained from `powerlaw_results.db` by the queries in `Manuscript/verification/`. The comparison columns are computed as

`R>0` ⇒ PL preferred; `R>0 AND p<0.05` ⇒ PL supported; `R<0 AND p<0.05` ⇒ exponential supported.

Note that `year = 0` in the `episcanner` tables is a pooled all-years fit, not a calendar year, and is excluded from all year counts; the national levels therefore have 15 annual fits, not 16.

References

Araujo, E. C., Codeço, C. T., Loch, S., Vacaro, L. B., Freitas, L. P., Lana, R. M., Bastos, L. S., de Almeida, I. F., Valente, F., Carvalho, L. M., et al. (2025). Large-scale epidemiological modelling: scanning for mosquito-borne diseases spatio-temporal patterns in brazil. *Royal Society Open Science*, 12(5):241261.

Becker, N. G. and Britton, T. (1999). Statistical studies of infectious disease incidence. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 61(2):287–307.

Clauset, A., Shalizi, C. R., and Newman, M. E. (2009). Power-law distributions in empirical data. *SIAM review*, 51(4):661–703.

Li, Y., Brown, P., Gesink, D. C., and Rue, H. (2012). Log gaussian cox processes and spatially aggregated disease incidence data. *Statistical methods in medical research*, 21(5):479–507.

Neipel, J., Bauermann, J., Bo, S., Harmon, T., and Jülicher, F. (2020). Power-law population heterogeneity governs epidemic waves. *PloS one*, 15(10):e0239678.

Rhodes, C. J., Jensen, H. J., and Anderson, R. M. (1997). On the critical behaviour of simple epidemics. *Proceedings of the Royal Society of London. Series B: Biological Sciences*, 264(1388):1639–1646.

Siettos, C. I. and Russo, L. (2013). Mathematical modeling of infectious disease dynamics. *Virulence*, 4(4):295–306.